

Proxy

Sub-techniques (4)

Adversaries may use a connection proxy to direct network traffic between systems or act as an intermediary for network communications to a command and control server to avoid direct connections to their infrastructure. Many tools exist that enable traffic redirection through proxies or port redirection, including [HTRAN](#), ZXProxy, and ZXPortMap. ^[1] Adversaries use these types of proxies to manage command and control communications, reduce the number of simultaneous outbound network connections, provide resiliency in the face of connection loss, or to ride over existing trusted communications paths between victims to avoid suspicion. Adversaries may chain together multiple proxies to further disguise the source of malicious traffic.

Adversaries can also take advantage of routing schemes in Content Delivery Networks (CDNs) to proxy command and control traffic.

ID: T1090

Sub-techniques: [T1090.001](#), [T1090.002](#), [T1090.003](#), [T1090.004](#)

①

Tactic: [Command and Control](#)

①

Platforms: ESXi, Linux, Network Devices, Windows, macOS

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[Version Permalink](#)

Procedure Examples

ID	Name	Description
C0063	2025 Poland Wiper Attacks	During the 2025 Poland Wiper Attacks , the adversaries utilized the rsocx tool identified as <code>r.exe</code> and <code>rsocx.exe</code> to tunnel within the internal infrastructure using a Reverse SOCKS Proxy. ^{[2][3]}
G0096	APT41	APT41 used a tool called CLASSFON to covertly proxy network communications. ^[4]
S0456	Aria-body	Aria-body has the ability to use a reverse SOCKS proxy module. ^[5]
S0347	AuditCred	AuditCred can utilize proxy for communications. ^[6]
S0245	BADCALL	BADCALL functions as a proxy server between the victim and C2 server. ^[7]
S1081	BADHATCH	BADHATCH can use SOCKS4 and SOCKS5 proxies to connect to actor-controlled C2 servers. BADHATCH can also emulate a reverse proxy on a compromised machine to connect with actor-controlled C2 servers. ^[8]
S0268	Bisonal	Bisonal has supported use of a proxy server. ^[9]
G0108	Blue Mockingbird	Blue Mockingbird has used FRP , ssf, and Venom to establish SOCKS proxy connections. ^[10]
C0017	C0017	During C0017 , APT41 used the Cloudflare CDN to proxy C2 traffic. ^[11]
C0027	C0027	During C0027 , Scattered Spider installed the open-source rsocx reverse proxy tool on a targeted ESXi appliance. ^[12]
S0348	Cardinal RAT	Cardinal RAT can act as a reverse proxy. ^[13]
G1021	Cinnamon Tempest	Cinnamon Tempest has used a customized version of the lox port-forwarding and proxy tool. ^[14]
G1052	Contagious Interview	Contagious Interview has leveraged Astrill VPN for C2. ^[15]
G0052	CopyKittens	CopyKittens has used the AirVPN service for operational activity. ^[16]
S0384	Dridex	Dridex contains a backconnect module for tunneling network traffic through a victim's computer. Infected computers become part of a P2P botnet that can relay C2 traffic to other infected peers. ^{[17][18]}
G1006	Earth Lusca	Earth Lusca adopted Cloudflare as a proxy for compromised servers. ^[19]
G0117	Fox Kitten	Fox Kitten has used the open source reverse proxy tools including FRPC and Go Proxy to establish connections from C2 to local servers. ^{[20][21][22]}
S1144	FRP	FRP can proxy communications through a server in public IP space to local servers located behind a NAT or firewall. ^[23]
S1044	FunnyDream	FunnyDream can identify and use configured proxies in a compromised network for C2 communication. ^[24]
G0047	Gamaredon Group	Gamaredon Group has used the Cloudflare Tunnel client to proxy C2 traffic. ^[25]
S1197	GoBear	GoBear implements SOCKS5 proxy functionality. ^[26]
S0690	Green Lambert	Green Lambert can use proxies for C2 traffic. ^{[27][28]}

ID	Name	Description
S0376	HOPLIGHT	HOPLIGHT has multiple proxy options that mask traffic between the malware and the remote operators. ^[31]
S0040	HTRAN	HTRAN can proxy TCP socket connections to obfuscate command and control infrastructure. ^{[32][33]}
S0283	jRAT	jRAT can serve as a SOCKS proxy server. ^[34]
S9044	Kali365	Kali365 has leveraged Cloudflare workers as reverse proxy infrastructure. ^{[35][36][37]}
S1190	Kapeka	Kapeka can identify system proxy settings via <code>WinHttpGetIEProxyConfigForCurrentUser()</code> during initialization and utilize these settings for subsequent command and control operations. ^[38]
S0487	Kessel	Kessel can use a proxy during exfiltration if set in the configuration. ^[39]
S1051	KEYPLUG	KEYPLUG has used Cloudflare CDN associated infrastructure to redirect C2 communications to malicious domains. ^[11]
S0669	KOCTOPUS	KOCTOPUS has deployed a modified version of Invoke-Ngrok to expose open local ports to the Internet. ^[40]
G1004	LAPSUS\$	LAPSUS\$ has leverage NordVPN for its egress points when targeting intended victims. ^[41]
S1121	LITTLELAMB.WOOLTEA	LITTLELAMB.WOOLTEA has the ability to function as a SOCKS proxy. ^[42]
S1141	LunarWeb	LunarWeb has the ability to use a HTTP proxy server for C&C communications. ^[43]
G0059	Magic Hound	Magic Hound has used Fast Reverse Proxy (FRP) for RDP traffic. ^[44]
G1054	MirrorFace	MirrorFace has used the GO Simple Tunnel (GOST) proxy tool. ^[45]
G1019	MoustachedBouncer	MoustachedBouncer has used a reverse proxy tool similar to the GitHub repository revsocks. ^[46]
G0069	MuddyWater	MuddyWater has used NordVPN to proxy phishing emails, making them appear to originate from France. ^[47]
S1189	Neo-reGeorg	Neo-reGeorg has the ability to establish a SOCKS5 proxy on a compromised web server. ^[48]
S0108	netsh	netsh can be used to set up a proxy tunnel to allow remote host access to an infected host. ^[49]
S0198	NETWIRE	NETWIRE can implement use of proxies to pivot traffic. ^[50]
S0508	ngrok	ngrok can be used to proxy connections to machines located behind NAT or firewalls. ^{[51][52]}
C0048	Operation MidnightEclipse	During Operation MidnightEclipse , threat actors used the GO Simple Tunnel reverse proxy tool. ^[53]
C0013	Operation Sharpshooter	For Operation Sharpshooter , the threat actors used the ExpressVPN service to hide their location. ^[54]
C0014	Operation Wocao	During Operation Wocao , threat actors used a custom proxy tool called "Agent" which has support for multiple hops. ^[55]
S0435	PLEAD	PLEAD has the ability to proxy network communications. ^[56]
S0007	Pulsedive	Pulsedive has used a proxy tool to connect to victims. ^[57]

ID	Name	Description
S1212	RansomHub	RansomHub can use a proxy to connect to remote SFTP servers. ^[61]
C0047	RedDelta Modified PlugX Infection Chain Operations	Mustang Panda proxied communication through the Cloudflare CDN service during RedDelta Modified PlugX Infection Chain Operations . ^[62]
C0056	RedPenguin	During RedPenguin , UNC3886 used malware capable of establishing a SOCKS proxy connection to a specified IP and port. ^{[63][64]}
S1187	reGeorg	reGeorg can establish an HTTP or SOCKS proxy to tunnel data in and out of a network. ^{[65][66][67]}
S0332	Remcos	Remcos uses the infected hosts as SOCKS5 proxies to allow for tunneling and proxying. ^{[68][69]}
S1210	Sagerunex	Sagerunex uses several proxy configuration settings to ensure connectivity. ^[70]
C0059	Salesforce Data Exfiltration	During Salesforce Data Exfiltration , threat actors used Mullvad VPN IPs to proxy voice phishing calls. ^[71]
S1099	Samurai	Samurai has the ability to proxy connections to specified remote IPs and ports through a a proxy module. ^[72]
G0034	Sandworm Team	Sandworm Team 's BCS-server tool can create an internal proxy server to redirect traffic from the adversary-controlled C2 to internal servers which may not be connected to the internet, but are interconnected locally. ^[73]
G1015	Scattered Spider	Scattered Spider has used proxy networks to hamper detection and has installed legitimate proxy tools on VMware vCenter and adversary-controlled VMs. ^{[74][75]}
S0461	SDBbot	SDBbot has the ability to use port forwarding to establish a proxy between a target host and C2. ^[76]
C0058	SharePoint ToolShell Exploitation	During SharePoint ToolShell Exploitation , threat actors used Fast Reverse Proxy to communicate with C2. ^{[77][78]}
S0273	Socksbot	Socksbot can start SOCKS proxy threads. ^[79]
S0615	SombRAT	SombRAT has the ability to use an embedded SOCKS proxy in C2 communications. ^[80]
S0436	TSCookie	TSCookie has the ability to proxy communications with command and control (C2) servers. ^[81]
G0010	Turla	Turla RPC backdoors have included local UPnP RPC proxies. ^[82]
S0263	TYPEFRAME	A TYPEFRAME variant can force the compromised system to function as a proxy server. ^[83]
S0386	Ursnif	Ursnif has used a peer-to-peer (P2P) network for C2. ^{[84][85]}
S0207	Vasport	Vasport is capable of tunneling though a proxy. ^[86]
G1017	Volt Typhoon	Volt Typhoon has used compromised devices and customized versions of open source tools such as FRP (Fast Reverse Proxy), Earthworm, and Impacket to proxy network traffic. ^{[87][88][89]}
S0670	WarzoneRAT	WarzoneRAT has the capability to act as a reverse proxy. ^[90]
G0124	Windigo	Windigo has delivered a generic Windows proxy Win32/Glubiteta.M. Windigo has also used multiple reverse proxy chains as part of their C2 infrastructure. ^[91]

Mitigations

ID	Mitigation	Description
M1037	Filter Network Traffic	Traffic to known anonymity networks and C2 infrastructure can be blocked through the use of network allow and block lists. It should be noted that this kind of blocking may be circumvented by other techniques like Domain Fronting .
M1031	Network Intrusion Prevention	Network intrusion detection and prevention systems that use network signatures to identify traffic for specific adversary malware can be used to mitigate activity at the network level. Signatures are often for unique indicators within protocols and may be based on the specific C2 protocol used by a particular adversary or tool, and will likely be different across various malware families and versions. Adversaries will likely change tool C2 signatures over time or construct protocols in such a way as to avoid detection by common defensive tools. ^[96]
M1020	SSL/TLS Inspection	If it is possible to inspect HTTPS traffic, the captures can be analyzed for connections that appear to be domain fronting.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0445	Detection of Proxy Infrastructure Setup and Traffic Bridging	AN1229	Suspicious process spawning (e.g., <code>rundll32</code> , <code>svchost</code> , <code>powershell</code> , or <code>netsh</code>) followed by network connection creation to internal hosts or uncommon external endpoints on high or non-standard ports.
		AN1230	User-space tools (e.g., <code>socat</code> , <code>ncat</code> , <code>iptables</code> , <code>ssh</code>) used in non-standard ways to establish reverse shells, port-forwarding, or inter-host connections. Often chained with uncommon outbound destinations or SSH tunnels.
		AN1231	AppleScript, LaunchAgents, or remote login services (<code>ssh</code> , <code>networksetup</code>) establishing proxy tunnels or dynamic port forwards to external IPs or alternate local hosts.
		AN1232	Direct use of <code>nc</code> , <code>socat</code> , or reverse tunnel scripts initiated by abnormal user contexts or unauthorized VIBs initiating connections from hypervisor to external systems.
		AN1233	Dynamic or static port forwarding rules added to route traffic through an internal host, or configuration changes to proxy firewall rules not aligned with baselined policy.

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